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# Juyi Lin

lin.juyi@northeastern.edu | (+1) 617-212-8191 | Northeastern University, Boston, US  
GitHub | LinkedIn | Homepage | Google Scholar

## EDUCATION

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**PhD student at Northeastern University, Boston, US** Sept 2024 - Now  
Major in Computer Engineering, GPA: 3.92/4, Advisor: Prof. Yanzhi Wang  
**King Abdullah University of Science and Technology** Sept.2022 - May.2024  
Master of Science in Computer Science, GPA: 3.71/4, Advisor: Marco Canini  
**Zhejiang University** Sept. 2018 - June 2022  
Bachelor of Engineering in Electronic Engineering, GPA: 3.75/4

## EXPERIENCES

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**Research Internship** May 2026 - August 2026  
Futurewei Technologies, Inc.  

- Research Intern, Generative AI & World Modeling team.
- Built **LEAP**, a coarse-to-fine perception router on top of **Qwen3 30BA3B omni model**, improving the watch-the-whole-clip baseline by **+6.0 pt** on OmniVideoBench, and staying +3 pt overall across 5 public long-video QA benchmarks.

**Infrastructure Internship** March 2022 - July 2022  
Beijing VirtAI Tech  

- Integrated PaddlePaddle distribution library and Deep Graph Library(DGL) into the performance measurement system.
- Conducted an in-depth performance analysis of PaddlePaddle distributed training.

## PROJECTS

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**Physical Reasoning Benchmark for Generative World Models** Sept 2025 – Mar 2026  
NEU  

- Proposed and built **PhyGround**, a criteria-grounded benchmark for physical reasoning in generative video world models across a taxonomy of **13 physical laws** (solid-body mechanics, fluid dynamics, optics), evaluating 8 video models. Paper: link. Code: link. Website: link. HF: link.
- Led a large-scale human annotation study involving **459 annotators**, **5,796 complete annotations**, **37.4K+ fine-grained labels**, with split-half ranking correlation Spearman's  $\rho > 0.90$ .
- Trained and released **PhyJudge-9B**, an open physics-specialized VLM judge that achieves **3.3% aggregate relative bias vs. 16.6% for Gemini-3.1-Pro**.

**Video Reasoning Benchmark** Sept 2025 – Jan 2026  
NEU  

- Proposed and implemented VBVR-EvalKit, a benchmark for evaluating reasoning capabilities in video generation models, covering 40 models. Paper: link Code: link. Website: link. HF: link.
- Co-lead a team of 22 researchers, designing the data generation and benchmark framework, defining the development roadmap, and contributing core code and evaluation metrics. Launched an open-source ecosystem including 100+ synthetic data generators.
- Proposed and built a video reasoning dataset, covering 200 curated reasoning tasks and 1M+ video clips (1000× larger than prior datasets).

**Vision Language Action(VLA) Model** Jan 2025 – Sept 2025  
NEU & Funded by EmbodysX  

- Proposed **VOTE**, an efficient fine-tuning framework for parallel action prediction in VLA models, reducing computational overhead and accelerating inference. Used by Cisco and EmbodysX. Paper link: VOTE. Code Link: Github.
- Proposed an ensemble voting strategy for the action sampling, improving model performance and enhances generalization across diverse tasks.

- Improved the average success rates of OpenVLA by over 20% across four LIBERO task suites, surpassed 7% average success rate of the state-of-the-art VLA model in SimplerEnv WidowX Robot, and accelerated action generation throughput by 39× on edge device NVIDIA Jetson Orin.
- Pretrained Llama3.2-1B LLM into a vision-language model (VLM) and fine-tuned with VOTE method, surpassing 7B OpenVLA’s average success rate by 16% across four LIBERO task suites.

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## SELECTED PUBLICATIONS

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The \* sign denotes equal contribution.

PhyGround: Benchmarking Physical Reasoning in Generative World Models

*Preprint*

**Juyi Lin**, Arash Akbari, Yumei He, Lin Zhao, Haichao Zhang, Arman Akbari, Xingchen Xu, Zoe Y. Lu, Enfu Nan, Hokin Deng, Edmund Yeh, Sarah Ostadabbas, Yun Fu, Jennifer Dy, Pu Zhao, Yanzhi Wang

VOTE: Vision-Language-Action Optimization with Trajectory Ensemble Voting

*ICANN 2026*

**Juyi Lin**, Amir Taherin, Arash Akbari, Arman Akbari, Lei Lu, Guangyu Chen, Taskin Padir, Xiaomeng Yang, Weiwei Chen, Yiqian Li, Xue Lin, David Kaeli, Pu Zhao, Yanzhi Wang

A Very Big Video Reasoning Suite

*ICML 2026*

Maijunxian Wang\*, Ruisi Wang\*, **Juyi Lin**\*, Ran Ji\*, Thaddäus Wiedemer, Qingying Gao, and others

Cross-Platform Scaling of Vision-Language-Action Models from Edge to Cloud GPUs

*GLSVLSI 2026*

Amir Taherin, **Juyi Lin**, Arash Akbari, Arman Akbari, Pu Zhao, Weiwei Chen, David Kaeli, Yanzhi Wang

ALTER: All-in-One Layer Pruning and Temporal Expert Routing for Efficient Diffusion Generation

*NeurIPS 2025*

Xiaomeng Yang\*, Lei Lu\*, Qihui Fan, Changdi Yang, **Juyi Lin**, Yanzhi Wang, Xuan Zhang, Shangqian Gao

**Open-Source Acceleration of Stable-Diffusion.cpp Deployable on All Devices**

*Preprint*

Jingxu Ng, Cheng Lv, Pu Zhao, Wei Niu, **Juyi Lin**, Minzhou Pan, Yun Liang, Yanzhi Wang

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## PUBLIC SERVICES

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- ICANN-26 Reviewer, AAAI-26/27 Program Committee, EuroSys’23 Shadow PC Reviewer
- Artifact Evaluation: MLSys’23, EuroSys’24/25, OSDI’24 & ATC’24
- Open Source Promotion Plan 2023

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## SKILLS

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**Programming Languages:** Python, C++, Golang, Bash, Matlab, SQL, C

**Tools:** Git, LaTeX, Docker

**Libraries/Frameworks:** PyTorch, LoRA, Transformers, Diffusers, Opencompass, LLaMA-Factory, ms-swift, Numpy